Gradient Descent Manual Calculations

Given:

$$\hat{y} = Xm + b$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \qquad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$m^{(0)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \qquad b^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$J(m,b) = \frac{1}{n}(\hat{y}-y)^T(\hat{y}-y)$$

$$m_{new} = m_{old} - \alpha \frac{\partial J}{\partial m}$$

$$b_{new} = b_{old} - \alpha \frac{\partial J}{\partial b}$$

Gradients:

With respect to m: $u = \hat{y} - y = Xm + b - y$

$$J = \frac{1}{n}\sum_{i=1} u_i^2$$

$$\frac{\partial J}{\partial u_i} = \frac{2}{n}u_i, \quad \frac{\partial u_i}{\partial m_j} = x_{ij}$$

By chain rule: $\frac{\partial J}{\partial m_j} = \sum_i \frac{\partial J}{\partial u_i}\frac{\partial u_i}{\partial m_j} = \frac{2}{n}\sum_i u_i x_{ij}$

$$\Rightarrow \frac{\partial J}{\partial m} = \frac{2}{n}X^T u$$

With respect to b: $\frac{\partial u_i}{\partial b_k} = \begin{cases} 1 & k=i \\ 0 & k \neq i \end{cases}$

$$\Rightarrow \frac{\partial J}{\partial b} = \frac{2}{n}(\hat{y}-y)$$

$n = 2$ so factor $\frac{2}{n} = 1$

$$\therefore \boxed{\frac{\partial J}{\partial m} = X^T(\hat{y}-y)} \qquad \therefore \boxed{\frac{\partial J}{\partial b} = \hat{y}-y}$$

Scanned with CamScanner

Scanned with CamScanner

Update 1 : done by : William MUKAMUKIZA

First : $m = [-1.0000, 2.0000]^T$, $b = [1.0000, 1.0000]^T$

Step 1 - Predictions :

$\hat{y} = Xm + b$

$\hat{y}_1 = 1 \times (-1.0000) + 3 \times 2.0000 + 1.0000 = \boxed{6.0000}$

$\hat{y}_2 = 4 \times (-1.0000) + 10 \times 2.0000 + 1.0000 = \boxed{17.000}$

Step 2 : $u = \hat{y} - y$

$u_1 = 6.0000 - 5 = \boxed{1.0000}$

$u_2 = 17.0000 - 6 = \boxed{11.0000}$

Step 3 : - Cost : $J = \frac{1}{2}(u_1^2 + u_2^2)$

$J = \frac{1}{2}[(1.0000)^2 + (11.0000)^2] = \boxed{61.0000}$

Step 4 : Gradient with respect to m $\partial J / \partial m = \frac{2}{2} X^T u = X^T u$

$\frac{\partial J}{\partial m_1} = 1 \times 1.0000 + 4 \times 11.0000 = \boxed{45.0000}$

$\frac{\partial J}{\partial m_2} = 3 \times 1.0000 + 10 \times 11.0000 = \boxed{113.0000}$

Step 5 : Gradient with respect to b $\partial J / \partial b = \frac{2}{2} u = u$

$\frac{\partial J}{\partial b_1} = u_1 = \boxed{1.0000}$, $\boxed{\frac{\partial J}{\partial b} = u_2 = 11.0000}$

Step 6 : Update : $\alpha = 0.01$ (learning rate) $= old - \alpha \times gradient$

$m_1 = -1.0000 - 0.01 \times 45.0000 = -1.0000 - 0.45000 = \boxed{-1.4500}$

$m_2 = 2.0000 - 0.01 \times 113.0000 = 2.0000 - 1.13000 = \boxed{0.8700}$

$b_1 = 1.0000 - 0.01 \times 1.0000 = 1.0000 - 0.01000 = \boxed{0.9900}$

$b_2 = 1.0000 - 0.01 \times 11.0000 = 1.0000 - 0.11000 = \boxed{0.8900}$

After Update 1 : $m = [-1.4500, 0.8700]^T$

Cost before this Update : 61.0000

Scanned with CamScanner

Scanned with CamScanner

Scanned with CamScanner

Mo Tu We Th Fr Sa Su

MEMO No. _____

DATE / /

Update 2 done by : Bedine Mukabucyana

$m = (-1.4500, 0.8700)^T$, $b = (0.9900, 0.89)^T$

write the formula, then each row

$\hat{y} = Xm + b$

$\hat{y}_1 = 1(-1.45) + 3 \times 0.8700 + 0.99 = 2.15$

$\hat{y}_2 = 4(1.45) + 10 \times 0.87 + 0.89 = 3.79$

step 2: Errors

$U = \hat{y} - y$

$U_1 = 2.15 - 5 = -2.85$

$U_2 = 3.79 - 6 = -2.21$

step 3: Cost

$J = \frac{1}{2}(U_1^2 + U_2^2)$

$J = \frac{1}{2}\left[(-2.85)^2 + (-2.21)^2\right]$

$= \frac{1}{2}(8.1225 + 4.8841)$

$= 6.5033$

Scanned with CamScanner

Scanned with CamScanner

Scanned with CamScanner

Mo Tu We Th Fr Sa Su

MEMO No. _____

DATE / /

step 4: Gradient with respect to m

$\partial J / \partial m = \frac{2}{2} X^T u = X^T u$

$\frac{\partial J}{\partial m_1} = 1(-2.85) + 4 \times (-2.21) = -11.69$

$\frac{\partial J}{\partial m_2} = 3(-2.85) + 10 \times (-2.21) = -30.65$

step 5: Gradient with respect to b

$\frac{\partial J}{\partial b} = \frac{2}{2} u = u$

$\frac{\partial J}{\partial b_1} = u_1 = -2.85$

$\frac{\partial J}{\partial b_2} = u_2 = -2.21$

step 6: Update $\alpha = 0.01$ (new = old - α x gradient)

$m_1 = -1.45 - 0.01(-11.69)$

$= -1.45 - (-0.1169)$

$= 1.1785 - 1.3331$

Scanned with CamScanner

Scanned with CamScanner

Scanned with CamScanner

Mo Tu We Th Fr Sa Su

MEMO No. _____

DATE / /

$m_2 = 0.8700 - 0.01 \times (-30.65)$

$= 0.87 - (-0.3065)$

$= 1.1765$

$b_1 = 0.99 = 0.01(-2.85)$

$= 0.99 - (-0.0285$

$= 1.0185$

$b_2 = 0.89 - 0.01(-2.21)$

$= 0.89 - (-0.0221)$

$= 0.9121$

After Update 2 :

$m = (-1.3331, 1.1765)^T$

$b = (1.0185, 0.9121)^T$

Cost before this update : 6.5033

Scanned with CamScanner

Scanned with CamScanner

Scanned with CamScanner

Update 3: Done by: Cedric Hinalocan

$m = (-1.3331, 1.1765)^T$, $b = (1.0195, 0.9121)^T$

prediction: $\hat{y} = Xm + b$

$\hat{y}_1 = 1 \times (-1.3331) + 3 \times (1.1765) + 1.0195 = \boxed{3.2149}$

$\hat{y}_2 = 4(-1.3331) + 10(1.1765) + 0.9121 = \boxed{7.3447}$

Error: $u = \hat{y} - y$

$u_1 = 3.2149 - 5 = \boxed{-1.7851}$

$u_2 = 7.3447 - 6 = \boxed{1.3447}$

Cost: $J = \frac{1}{2}(u_1^2 + u_2^2)$

$= \frac{1}{2}\left[(-1.7851)^2 + (1.3447)^2\right]$

$= \frac{1}{2}(3.1866 + 1.8082)$

$= \frac{1}{2}(4.9948) = \boxed{2.4974}$

Gradient with respect to m_i:

$\frac{\partial J}{\partial m_1} = X^T u$

$\frac{\partial J}{\partial m_1} = 1(-1.7851) + 4(1.3447) = 3.5937$

$\frac{\partial J}{\partial m_2} = 3(-1.7851) + 10(1.3447) = 8.0917$

Gradient with respect to b_i:

$\frac{\partial J}{\partial b} = u$

$\therefore \frac{\partial J}{\partial b_1} = u_1 = -1.7851 \qquad \frac{\partial J}{\partial b_2} = u_2 = 1.3447$

Update: $\alpha = 0.01$

$m_1 = -1.3331 - 0.01(3.5937) = \boxed{-1.3690}$

$m_2 = 1.1765 - 0.01(8.0917) = 1.0956$

$b_1 = 1.0195 - 0.01(-1.7851) = 1.0364$

$b_2 = 0.9121 - 0.01(1.3447) = 0.8987$

$m = \begin{bmatrix} -1.3690 \\ \end{bmatrix} \qquad b = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$

Scanned with CamScanner

Result and trend

After 3 updates; $m = \begin{pmatrix} -1.3690 \\ 1.0956 \end{pmatrix}$, $b = \begin{pmatrix} 1.0364 \\ 0.8787 \end{pmatrix}$, $J = \begin{pmatrix} 1.9541 \\ 6.3393 \end{pmatrix}$

$\bar{J} = 2.1645$.

The cost falls at every step so the parameters are moving in the direction that reduces the error, as each step goes against the gradient. The first step is large because the initial gradient is big.

Scanned with CamScanner

Scanned with CamScanner